

K-audit — Substrate-SM Dynamics Engine Consolidation v0.1 (Elie #410); verdict CONDITIONAL PASS with 3 conditions

Keeper

2026-05-29 Friday EDT (post-Lyra #414 reframe)

K-audit — Dynamics Engine Consolidation v0.1

Verdict

CONDITIONAL PASS. The algebra is rigorous, the tier wall between RIGOROUS / BET / OPEN is held cleanly, the E1b retraction propagates correctly through the doc, and the consolidation honestly represents what's built vs. what isn't. Three items below required for v0.2 → clean PASS; none CRITICAL.

Audit hooks (Elie #410 §5) — addressed

1. E0 q-Serre $\leftrightarrow$ substrate-primary identification. VERIFIED. Gaussian q-integers at $q=2$ check: $[2]_2 = 1+2 = 3 = N_c$; $[3]_2 = 1+2+4 = 7 = g = M\{N_c\}$; at $q^2=4$: $[2]_4 = 1+4 = 5 = n_C$; $[3]_4 = 1+4+16 = 21 = N_c \cdot g$. Standard $U_q^+(B_2)$ Serre relations use the short-root q-integer and the long-root q^2 -integer (Cartan $a_{12} = -1$, $a_{21} = -2$), so these are the *defining* structure constants at $q=2$ — correct. The framing “not constant-by-constant derivation — the four primaries are the defining relations of one algebra” is accurate. PASS.
2. E2 Hall-number computation (A_2 slice). VERIFIED for the A_2 subquiver: $u_{S1} \cdot u_{S2} = \{E12:1, S1 \oplus S2:1\}$, $u_{S2} \cdot u_{S1} = \{S1 \oplus S2:1\}$, commutator = $\{E12:1\}$. Standard Hall arithmetic over $GF(2)$ (one non-split extension + one split = 2 modules; reversed product = split only). Commutator isolating the bound state is the canonical Hall $\rightarrow$ QG relation. PASS on the A_2 slice — see Finding 1 on the B_2 extension.
3. E3 grading-conservation argument. VERIFIED as algebra: a graded coproduct $\Delta: A_n \rightarrow \bigoplus_{i+j=n} A_i \otimes A_j$ automatically conserves any linear functional on the grading lattice (standard graded-bialgebra fact). The “3 independent charges $\Rightarrow$ rank-3 grading $\Rightarrow$ affine $\hat{B}_2$ ” argument is a genuine structural derivation — finite B_2 has $K_0 = \mathbb{Z}^2$, which can't host 3 independent functionals; affine $\hat{B}_2$ has $K_0 = \mathbb{Z}^3$, which can. This is real content. PASS on the structural argument — see Finding 3 on the gauge-charge count.
4. Tier wall. Audited each RIGOROUS claim against the BET; no RIGOROUS item secretly depends on the dictionary or the open gates. The algebra (multiplication, coproduct, grading, R-matrix, F-part) stands independently of “which module = which particle.” PASS.
5. Retraction propagation. §4 explicitly retracts E1b “3 tubes forced” and the doc carries no residual “forced=3” elsewhere. PASS.

Findings (severity-rated)

Finding 1 — MINOR

E2 is computed on the A_2 subquiver, not the full B_2 . The fusion arithmetic shown — $u_{S1} \cdot u_{S2} = \{E12, S1 \oplus S2\}$, commutator = $E12$ — is rigorous Hall arithmetic on the A_2 slice of B_2 . The substrate is B_2 , which has a *long-root* relation as well: the $[3]_4 = 21 = N_c \cdot g$ Serre identity, with three terms ($E2^3 E1$, $E2^2 E1 E2$, $E2 E1 E2^2$) and a higher Casimir cancellation. That long-root case isn't explicitly computed as a Hall product in §2. The full claim “fusion = Hall product on the substrate algebra” extrapolates the A_2 computation to all of B_2 by Ringel's theorem

(legitimate), but a worked B_2 long-root example would close the gap from *proof of concept* to *demonstrated on the substrate*. No correctness issue; completeness gap. Severity: MINOR.

Condition 1: v0.2 adds the long-root B_2 Hall computation (or a Cal cross-check on the long-root structure constants).

Finding 2 — MODERATE (*time-of-writing supersession*)

§4 generation disposition is now SUPERSEDED by Lyra #414 (filed later this session). Elie’s text reads “the discriminator is SM color-generation independence, which FAVORS (A) $h-1$.” That was the lean *before* Lyra’s #412 + #414 work, which (a) undercut route (A) — the δ -tube realization leans ≤ 2 , not 3 — and (b) reversed her own argument against route (II): color-blindness rules out generations *carrying* color, not the count *equaling* N_c via a colorless projection. The current Keeper disposition (locked in the honest-state ledger): route (I) UNDERCUT; favored mechanism shifts to (II) $h^\vee = N_c = \text{Grace’s Track P}$; count-NUMBER forced; generation-IDENTIFICATION mechanism OPEN with a two-structures burden. This isn’t an error in Elie’s doc — he wrote at 15:45 before #414 landed — but the audit doc must travel with the current disposition or it will mislead readers. Severity: MODERATE (load-bearing gate description).

Condition 2: v0.2 §4 absorbs the #414 reframe — route-(I) undercut, mechanism shifts to route-(II) Track P, count-number forced ($h^\vee = N_c = 3$), mechanism open with the burden of producing two independent 3-fold structures (colors AND generations) from one invariant. The earlier “two independent 3’s” framing is retired $\rightarrow$ “one 3 ($h^\vee$) double-duty.”

Finding 3 — MODERATE

The “3 independent charges $\Rightarrow$ rank-3 grading $\Rightarrow$ affine $\hat{B}_2$ ” argument may UNDERCOUNT the SM’s conserved-charge content. The argument is structurally sound (grading rank must accommodate independent conserved functionals), but (Q, B, L) is the *low-energy* count. At the gauge level the SM has $U(1)_Y \times SU(2)_L \times SU(3)_C$ with conserved hypercharge Y , weak isospin T_3 , and two color Cartan components — i.e., $1 + 1 + 2 = 4$ gauge Cartan charges, plus B and $L \rightarrow$ potentially more independent conserved functionals than rank-3 can host. Either (a) Q, B, L are genuinely the *fundamental* SM charges and the gauge Cartan reduces to them at the substrate level (needs argument), or (b) the substrate grading is richer than rank-3 (which would push past affine $\hat{B}_2$). The current text glides past this and reads as if “rank-3 $\Rightarrow$ affine $\hat{B}_2$ ” is forced without addressing the gauge count. Severity: MODERATE.

Condition 3: v0.2 §2 (E3) addresses the gauge-charge count — either argues why the SM Cartan reduces to (Q, B, L) at the substrate level (so rank-3 genuinely suffices), or notes the grading rank may need to extend (with the corresponding algebraic implication).

Net

The dynamics engine is genuinely built; “model the entire SM process” has a working, computed mechanism (fusion = product, decay = coproduct, conservation = grading, scattering = R-matrix, antimatter/CPT = F-part). Elie’s tier discipline is exemplary — every claim labeled, the bet isolated cleanly to the dictionary, the retraction owned and propagated. The three conditions above are honest *completions*, not corrections of errors. With them addressed (v0.2), this is a clean PASS and the load-bearing Goal-2 mechanism is K-audit-ratified.

— Keeper, K-audit verdict CONDITIONAL PASS, 2026-05-29.